

Dissection Protocol

Purpose:

Isolation of Nucleus Accumbens (NAc), cortex, and hippocampus.

Notes:

- Work quickly but gently — target ≤ 2 min per brain.
- Keep all tissues ice-cold at all times.

Materials:

- Tray
- Ice
- Ice Pack
- Dried Ice
- Whatman Filter Papers
- Scalpel
- Blade
- Spatulas
- Scissors
- Forceps
- Ice cold Phosphate Buffered Saline (PBS)
- Eppendorfs

Steps:

1. Set up your working area
 - Fill tray with ice
 - Top it with dry ice
 - Get ice pack and cover it with gloves, place it on top of tray.
 - On the ice pack, stack 2 Whatmans on the right and another 2 on the left.
 - Get PBS from fridge and a dropper
 - Garbage bag for trash
 - Put all the instruments you'll be using next to you.
2. Brain Isolation
 - Using scissors, cut middle incision the mouse head skin until the skull is exposed
 - Then, using scissors, middle incision the skull followed by two lateral incisions.
 - Now, using the forceps, trace your middle incision and pull away the skull sideways to expose the brain.
 - After the brain is exposed, using a spatula, scoop the brain out and place it on the second Whatmann filter paper.

3. Brain Dissection

- Using the dropper, add 4-5 drops of PBS to the brain
- Then, get the scalpel and make a horizontal incision at approximately 2/3 of the brain length anteriorly.
- Rotate that cut so you can see the inside of it.
- Now you can see two dots which are the Right NAc and the left NAc, using the spatulas, remove both NAc's.
- Next, to isolate the hippocampus and the Cortex, using the spatulas, scrape off the cortex in a direction middle to outside very gently, to reveal the hippocampus.
- You have now isolated the cortex, next remove the hippocampus which you have exposed.
- Repeat on the other side.
- Finally, you should have 2 NAc's, 2 cortexes, 2 hippocampi.
- Place each brain part in its corresponding previously labeled Eppendorf.